

Multiple Choice (10 marks)

1. Statement 1 | Every ideal in a ring is a subring of the ring. Statement 2 | Every subring of every ring is an ideal of the ring.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
2. Sofia and Tess will each randomly choose one of the 10 integers from 1 to 10. What is the probability that neither integer chosen will be the square of the other?
 - (a) 0.64
 - (b) 0.72
 - (c) 0.81
 - (d) 0.95
3. The polynomial $x^4 + 4$ can be factored into linear factors in $\mathbb{Z}_5[x]$. Find this factorization.
 - (a) $(x - 2)(x + 2)(x - 1)(x + 1)$
 - (b) $(x+1)^4$
 - (c) $(x-1)(x+1)^3$
 - (d) $(x-1)^3(x+1)$
4. Some group $(G, 0)$ is known to be abelian. Then which one of the following is TRUE for G ?
 - (a) $g = g^{-1}$ for every g in G
 - (a) $g = g^2$ for every g in G
 - (b) $(g \circ h)^2 = g^2 \circ h^2$ for every g, h in G
 - (c) G is of finite order
5. Suppose today is Wednesday. What day of the week will it be $10^{(10^{(10)})}$ days from now?
 6. Sunday
 7. Monday
 8. Tuesday
 9. Wednesday

True / False (10 marks)

Answer True or False

6. True or False: Every solvable group is of prime-power order, as solvable groups can have orders that are products of different primes.
7. True or False: The set of any two distinct column vectors of a 5×5 matrix M is linearly independent if M has a non-zero determinant.
8. True or False: The equation $\cos(97x) = x$ has exactly 31 positive solutions for x .
9. True or False: The dimension of the intersection of two 2-dimensional subspaces of R^4 can be 0, 1, or 2.
10. True or False: The probability of rolling a six 70 or more times in 360 tosses of a fair die is greater than 0.50.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Human beings have discovered a habitable planet and soon after, they find 10 more habitable planets. Of these 11, only 5 are deemed “Earth-like” in their resources and the rest are deemed “Mars-like” since they lack many important resources. Assume that planets like Earth take up 2 units of colonization, while those like Mars take up only 1. If humanity mobilizes 12 total units of colonies, how many different combinations of planets can be occupied if the planets are all different from each other?
12. What is the mean of the measures of the three exterior angles of a triangle if two of the interior angles have measures of 63 and 78 degrees?

End of Paper